

Pool boiling experiment characteristics on the pure copper surface

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Abstract

The pool boiling characteristics with different boiling surfaces and working fluids play an important part in multiphase flow research. The key parameters of pool boiling, such as heat transfer coefficient (HTC) and critical heat flux (CHF), can be only acquired by experiment. Thus, a pool boiling experimental device is designed and produces the HTC and CHF data on the pure copper heating surface, which are 72.25 kW/(m²·K) and 1093.28 kW/m², respectively. Besides, a series of visualization experimental results of bubble behavior in the pool boiling are taken by the high-speed camera to provide references for the boiling mechanism research. The pool boiling experiment would be the benchmark data to validate the future experiments and computer simulations.

Keywords

pool boiling
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1 Introduction

Boiling heat transfer is one of the most effective phase change heat transfer methods. It can utilize the huge latent heat of vaporization of the boiling liquid to transfer or dissipate high heat flux with a minimal temperature difference between the heat transfer surface and the liquid (Vakarelski et al., 2012; Cho et al., 2015, 2017). Thus, it plays an important part in heat energy transfer and conversion of industries, such as electricity generation, refrigeration equipment, electronics cooling, etc. The enhancement in boiling performance can bring immediate benefits to improve the efficiency, safety, and cost in energy systems, leading to energy savings and greenhouse gas emission reduction.

The two most important parameters in boiling heat transfer are boiling heat transfer coefficient (HTC) and critical heat flux (CHF) (Wang and Chen, 2018), which respectively characterized the heat transfer efficiency and the allowable heat transfer capacity. The CHF, a phenomenon also known as the boiling crisis (Lloveras et al., 2012), occurs accompanied by the sharp deterioration on the rate of

HTC from boiling surface to coolant (Yang et al., 2021), and leads to the sudden shift of flow regime from nucleate boiling to film boiling, which results in drastic increasing of surface temperature with disastrous consequences, for example, the degradation of surface material, the destruction of device components, and the thermal meltdown of the systems (Rahman et al., 2015; Jaikumar et al., 2017).

As the most basic one in the boiling experiments, the pool boiling experiment could be a feasible idea to get the key parameters of the boiling heat transfer and provide references for boiling mechanism research (Berenson, 1962; Dhillon et al., 2015). Besides, the pool boiling experiment would be the benchmark data for the computer simulations and other boiling experiments (McVay et al., 2015).

In this study, we get the HTC and CHF data on the pure copper surface from the pool boiling experimental device. In addition, the photos of bubbles at different heat flux have been captured by the high-speed camera.

2 Pool boiling experimental device

Figure 1 shows the diagram of the specially designed

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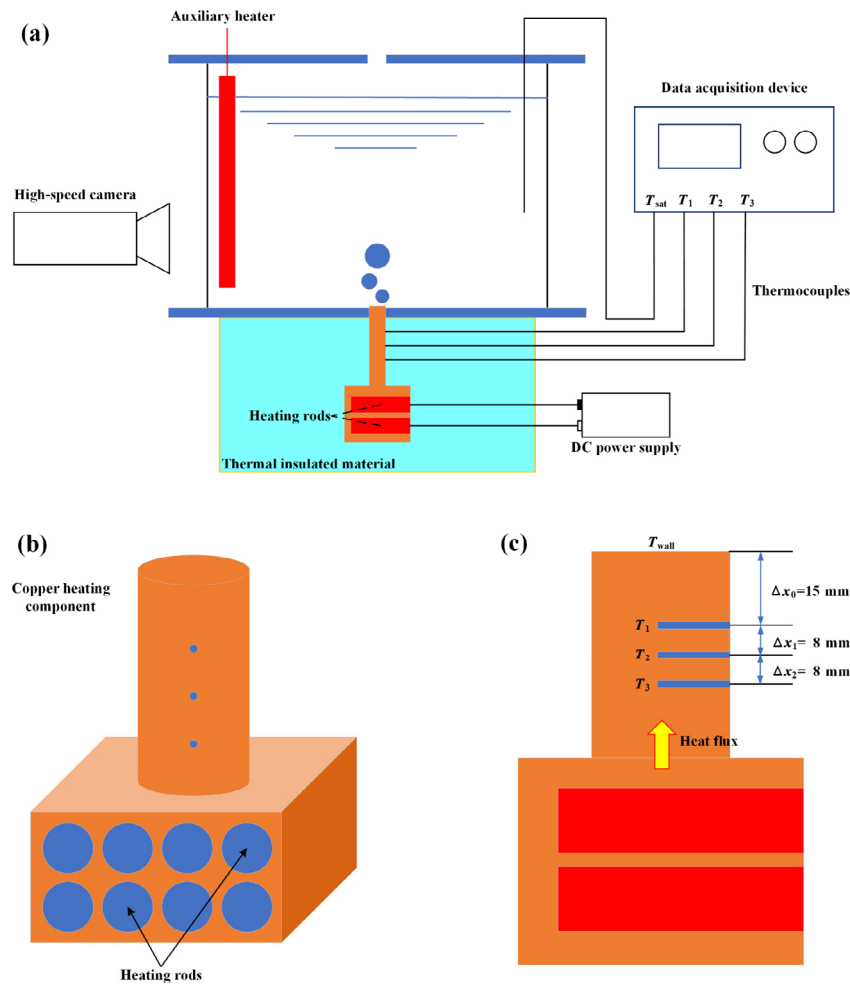


Fig. 1 Pool boiling experimental device: (a) schematic diagram of the pool boiling experimental device; (b) copper heating component; (c) K-type thermocouples' locations.

experimental device, which consists of a boiling pool chamber, heating component, auxiliary heater, data acquisition device, DC power supply, high-speed camera, and so on.

2.1 Boiling pool chamber and heating component

The overall dimensions of the boiling pool chamber are $160 \text{ mm} \times 120 \text{ mm} \times 150 \text{ mm}$. The wall of the chamber was made of 3 mm thick glass, while the bottom and top were respectively made of thermal insulated Teflon plate and FR4 fiberglass board. The boiling pool chamber was directly open to the atmosphere by the hole on the top plate, in order to keep the pool boiling experiment under atmospheric pressure.

Copper is one of the most widely used materials in scientific studies and engineering applications because of its excellent good processing properties and physical performance, such as high thermal conductivity and high electrical conductivity. As a result, copper is used as a base material in the current study. As shown in Fig. 1(b), the

heating component consisted of a copper component and eight heating rods ($\Phi 12 \text{ mm} \times 90 \text{ mm}$), which connected to the DC power supply (MAISHENG MP25020D DC power supply) with max power 5 kW (250 V, 20 A). The upper part of the copper component was a cylinder with 20 mm in diameter and 36 mm in height, while the bottom part was a cuboid with 90 mm in length, 65 mm in width, and 35 mm in height. The heating rods with max power 250 W were inserted into the bottom part of the copper component and directly provided heat flux to the heating component.

The upper part of the heating component passed through a hole located at the center of the Teflon bottom plate. A high-temperature silicone gasket was used to seal the gap between the heating component and the boiling pool chamber, in order to avoid the working fluid leakage of the experimental device and prevent boiling from occurring at the edge of the copper heating surface. Besides, the whole heating component was wrapped in the high-temperature resistance thermal insulated fiberglass

cotton blanket to reduce the heat loss to the environment. After solving the liquid leakage and heat loss, the distilled water, as working fluid, was heated and maintained at saturation temperature (T_{sat}) by a K-type thermocouple and the auxiliary heater (250 V, 250 W) placed inside the boiling pool chamber.

2.2 Experimental data acquisition and processing

Three sheathed K-type thermocouples (whose probe diameters were 1 mm, marked by T_1 , T_2 , and T_3) were installed inside three holes (diameter was 1 mm and depth was 11 mm) in the upper part of the heating component. As Fig. 1(c) shows, $\Delta x_0 = 15$ mm is the distance between the boiling surface and thermocouple T_1 , while $\Delta x_1 = 8$ mm is the distance between T_1 and T_2 , and $\Delta x_2 = 8$ mm is the distance between T_2 and T_3 . The thermal conductivity of pure copper (k_{Cu}) at 100–200 °C, which is about 389–393 W/(m·K), is large enough that the heat conduction through the upper copper cylinder can consider as one-dimension heat conduction. As a result, the heat flux (q''), the wall superheat (ΔT_{wall}), and the boiling heat transfer coefficient (h) can be calculated as Eqs. (1)–(3):

$$q'' = k_{\text{Cu}} \frac{\Delta T}{\Delta x} = k_{\text{Cu}} \left(\frac{\frac{T_3 - T_2}{\Delta x_2} + \frac{T_2 - T_1}{\Delta x_1}}{2} \right) \quad (1)$$

$$\Delta T_{\text{wall}} = T_{\text{wall}} - T_{\text{sat}} = \left(T_1 - \frac{q}{k_{\text{Cu}}} \Delta x_0 \right) - T_{\text{sat}} \quad (2)$$

$$h = \frac{q''}{\Delta T_{\text{wall}}} \quad (3)$$

The copper thermal conductivity k_{Cu} is approximately equal to 390 W/(m·K) as a constant.

The uncertainty of the temperature measurement, including T_1 , T_2 , T_3 , and T_{sat} , was determined by the accuracy of the K-type thermocouple, which was $U_{T_1} = U_{T_2} = U_{T_3} = U_{T_{\text{sat}}} = 0.5$ °C. The uncertainty of the distance measurement was determined by the Vernier caliper, which was $U_{\Delta x_0} = U_{\Delta x_1} = U_{\Delta x_2} = 0.02$ mm. As a result, the maximum relative uncertainties of the heat flux, the wall superheat, and the boiling heat transfer coefficient were calculated by the law of uncertainty propagation, which were $U_{q''}/q'' = 1.58\%$, $U_{\Delta T_{\text{wall}}}/\Delta T_{\text{wall}} = 6.43\%$, and $U_h/h = 6.62\%$, respectively. While the uncertainties were $U_{q'} = 17.32$ W/m², $U_{\Delta T_{\text{wall}}} = 0.97$ °C, and $U_h = 4.79$ W/(m²·K). The equations of uncertainty calculation are as Eqs. (4)–(6):

$$\frac{U_{q''}}{q''} = \sqrt{\left(\frac{k_{\text{Cu}}}{2} \right)^2 \times \left[\frac{U_{T_1}^2}{(\Delta x_1)^2} + \left(\frac{1}{\Delta x_1} - \frac{1}{\Delta x_2} \right)^2 U_{T_2}^2 + \frac{U_{T_3}^2}{(\Delta x_2)^2} + \left(\frac{T_2 - T_1}{(\Delta x_1)^2} \right)^2 U_{\Delta x_1}^2 + \left(\frac{T_3 - T_2}{(\Delta x_2)^2} \right)^2 U_{\Delta x_2}^2 \right]} \quad (4)$$

$$\frac{U_{\Delta T_{\text{wall}}}}{\Delta T_{\text{wall}}} = \sqrt{\frac{U_{T_1}^2 + \left(\frac{\Delta x_0}{k_{\text{Cu}}} \right)^2 U_{q''}^2 + \left(\frac{q''}{k_{\text{Cu}}} \right)^2 U_{\Delta x_0}^2 + U_{T_{\text{sat}}}^2}{\Delta T_{\text{wall}}^2}} \quad (5)$$

$$\frac{U_h}{h} = \sqrt{\frac{U_{q''}^2}{q''^2} + \frac{U_{\Delta T_{\text{wall}}}^2}{(\Delta T_{\text{wall}})^2}} \quad (6)$$

3 Pool boiling experiment on the pure copper surface

3.1 Experimental method

Before the experiment, the pure copper surface of the heating component was polished by 2000 mesh sandpaper and cleaned with 95% ethyl alcohol. Subsequently, the heating component was assembled inside the pool boiling experimental device and the formal experiment began. The main steps of a formal pool boiling experiment are as follows:

Step 1: The distilled water in the boiling pool chamber was heated by the auxiliary heater and maintained at saturation temperature (1 atm, 100 °C) for over 30 min, in order to eliminate the non-condensable gas in the working fluid.

Step 2: The heating component was switched on and kept at a steady-state condition, which is applied to calculate the heat flux and the boiling heat transfer performance. Each of the steady-state conditions was characterized by the maximum temperature fluctuations of T_1 , T_2 , and T_3 no more than 0.5 °C/min. The final steady-state data were the average of the temperature data collected by the data acquisition device (A-BF BCL2008S Multichannel temperature recorder) in 1 min. During the steady-state condition, photos should be taken by the high-speed camera to analyze the bubble behavior of the pool boiling.

Step 3: The heat flux supplied to the pure copper heating surface was stepwise increased by turning up the DC power supply, until the critical heat flux (CHF) occurred, whose phenomenon is not only the sharp increment of thermocouples temperature but also the film boiling on the heating surface.

Step 4: When the CHF phenomenon had occurred, the power source should be decreased or shut down before the heating component temperature is too high to break down the thermocouples or heating rods.

3.2 Results and discussion

Figures 2 and 3 and Tables 1 and 2 present the pool boiling performance on the pure copper surface in this work.

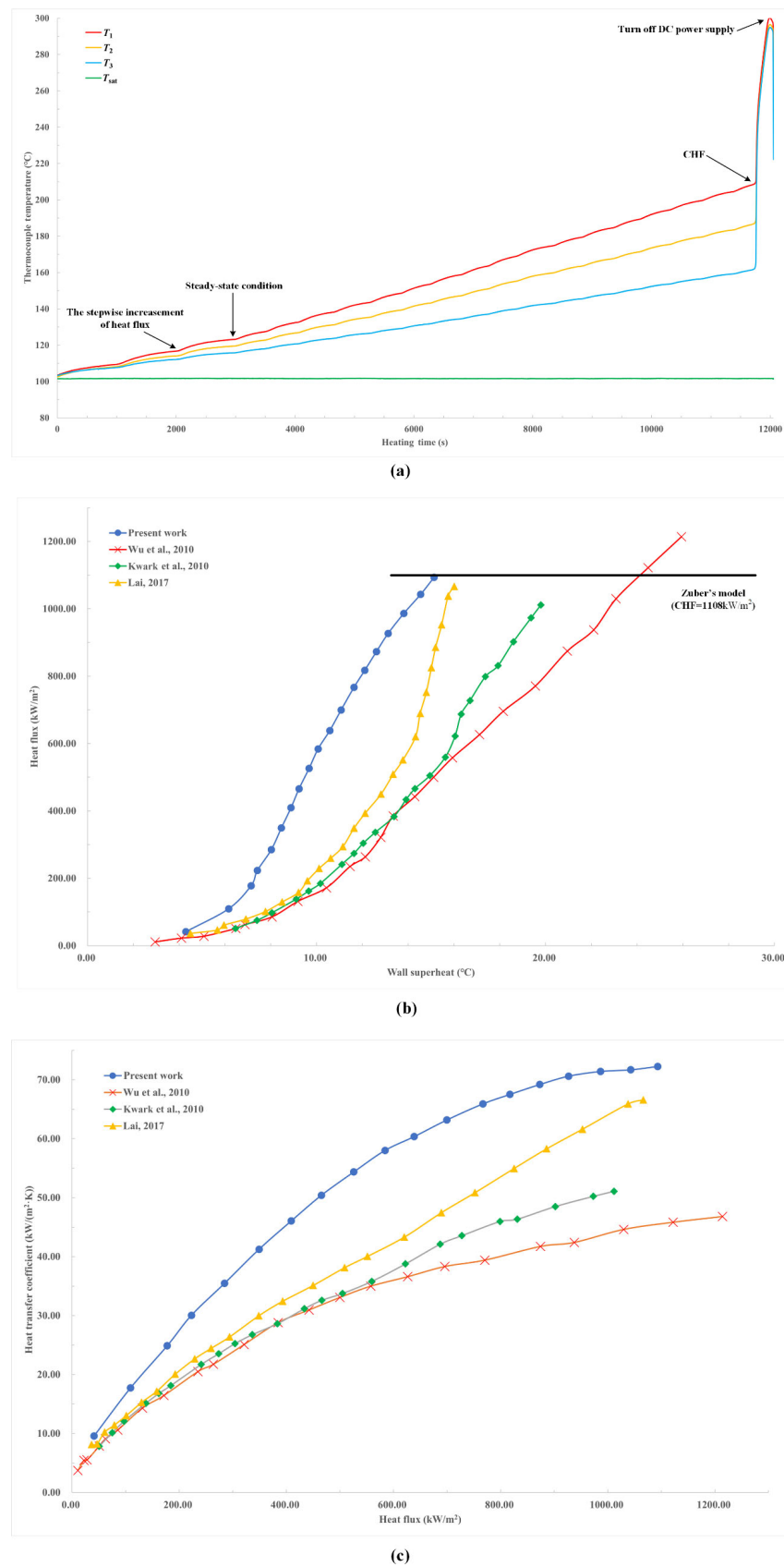


Fig. 2 Boiling heat transfer performance: (a) temperature curves of the thermocouples for the pure copper surface; (b) boiling curves for the pure copper surface in this work and other studies; (c) boiling heat transfer coefficients for the pure copper surface in this work and other studies.

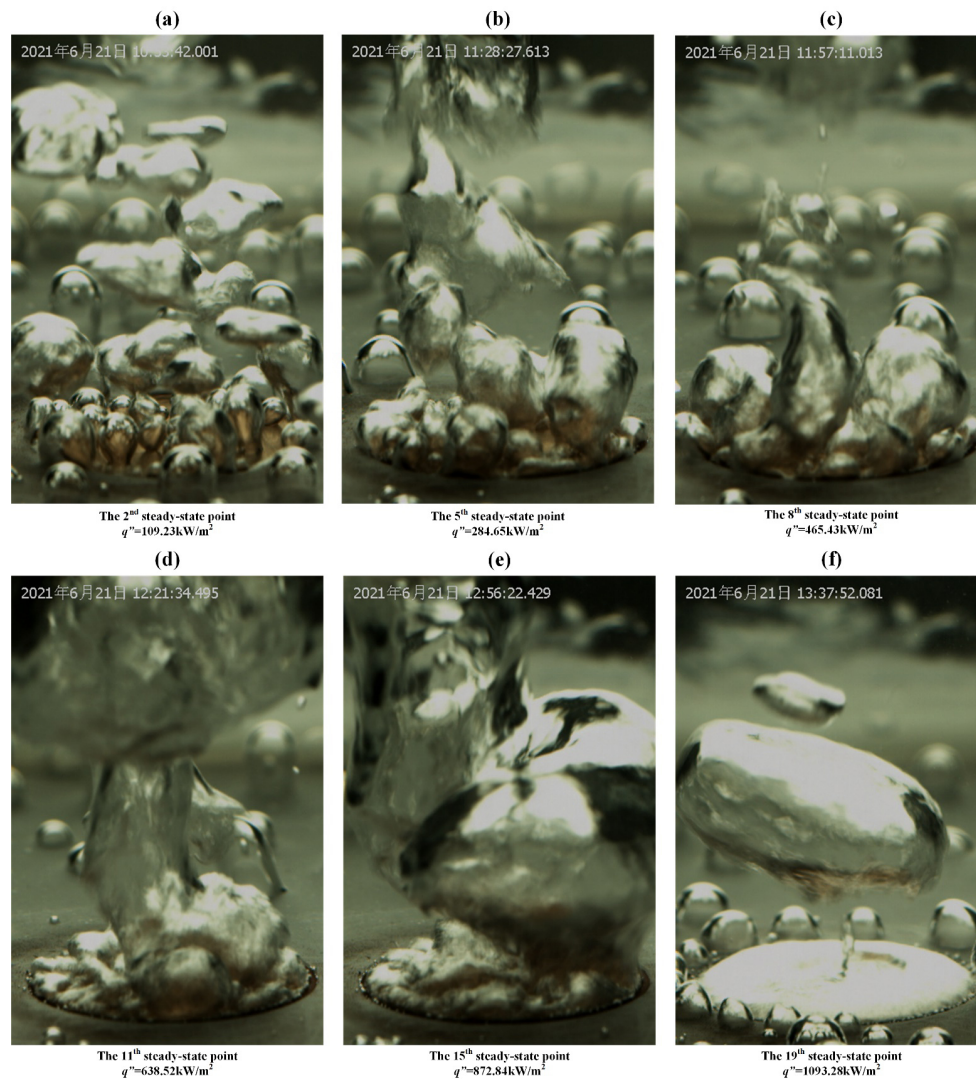


Fig. 3 Visualization experimental results of bubble coalescences on the pure copper surface under different steady-state heat flux.

Table 1 Comparison of the pool boiling experiments

	Material	Size of the heating surface	Heater	Working fluid
Present work	Copper	$\Phi 20$ mm	Heating rod	Water
Wu et al. (2010)	Copper	10 mm $\times$ 10 mm	Film resistive heater	Water
Kwark et al. (2010)	Copper	10 mm $\times$ 10 mm	Film resistive heater	Water
Lai (2017)	Copper	$\Phi 20$ mm	Heating rod	Water

When the CHF phenomenon is approached, the boiling surface is covered by a vapor film formed by vigorous coalescence between lots of bubbles on the heating surface, leading to a transition from nucleate boiling regime to film boiling regime. As a result, heat transfer between the heating surface and the working fluid is blocked by the vapor film, causing heat transfer deterioration which will characterize as sudden and dramatic wall temperature increment (Fig. 2(a)).

Figure 2(b) shows the comparison of the pool boiling in this work and other studies (Kwark et al., 2010; Wu et al., 2010; Lai, 2017). The obtained steady-state condition and CHF data are similar with both the experimental and theoretical results of pure surfaces in the other studies, verifying the accuracy and reliability of the current experimental device and method.

The same as the empirical trend, the HTC increases with the increase of heat flux, as shown in Fig. 2(c), but the

Table 2 Experimental results on the pure copper surface under different steady state

No.	T_1 (°C)	T_2 (°C)	T_3 (°C)	T_{sat} (°C)	q'' (kW/m ²)	T_{wall} (°C)	ΔT_{wall} (°C)	h (kW/(m ² ·K))
1	107.55	107.99	109.24	101.67	41.18	105.97	4.30	9.59
2	112.10	114.05	116.58	101.74	109.23	107.90	6.16	17.73
3	115.74	119.42	123.04	101.77	177.65	108.91	7.14	24.88
4	117.72	122.37	126.87	101.73	222.88	109.14	7.42	30.05
5	120.64	126.66	132.33	101.66	284.65	109.69	8.02	35.47
6	123.57	131.00	137.91	101.68	349.14	110.14	8.46	41.25
7	126.39	135.13	143.20	101.76	409.31	110.65	8.89	46.06
8	128.78	138.77	147.90	101.64	465.43	110.87	9.23	50.40
9	131.58	142.91	153.19	101.68	525.98	111.35	9.68	54.36
10	134.20	146.74	158.20	101.66	584.17	111.73	10.07	58.01
11	136.80	150.56	163.03	101.66	638.52	112.24	10.58	60.35
12	139.66	154.81	168.40	101.69	699.51	112.76	11.07	63.17
13	142.74	159.41	174.25	101.61	766.94	113.24	11.64	65.91
14	145.24	163.07	178.81	101.71	817.13	113.81	12.10	67.50
15	147.88	166.94	183.74	101.70	872.84	114.31	12.62	69.19
16	150.48	170.76	188.55	101.71	926.70	114.84	13.13	70.60
17	153.47	175.13	193.99	101.72	986.35	115.53	13.81	71.41
18	156.34	179.26	199.17	101.69	1042.64	116.24	14.54	71.69
19	158.88	182.90	203.80	101.70	1093.28	116.83	15.13	72.25

rate of increment somewhat decreases at the high heat flux. The HTC reaches the peak when the CHF phenomenon occurs.

All of the steady-state condition data are shown in Table 2 and the corresponding visualization experimental results are shown in Fig. 3. From visual inspection through high-speed CCD, at the beginning of the experiment, the bubbles are small and independent (Fig. 3(a)). With the rising of the heat flux, the small bubbles gather together and grow larger on the copper surface (Figs. 3(b) and 3(c)). The larger the heat flux rose, the growth and enlargement of localized hot/dry regions are easier to form large vapor slugs (Figs. 3(d) and 3(e)). The vapor slugs and bubbles will vigorously coalesce even further when the heat flux keeps increasing, until the formation of a vapor blanket that covers the boiling surface, which means a boiling regime transition (Fig. 3(f)).

4 Conclusions

In this work, the recent saturated pool boiling experimental results were introduced, which were conducted at atmospheric pressure using the pure copper as heating surface and the distilled water as working fluid. The HTC is 72.25 kW/(m²·K) and the CHF is 1093.28 kW/m², which were obtained by the specially designed pool boiling experimental

device. Furthermore, the visualization experimental results of pool boiling, such as typical nucleate boiling and typical film boiling, have been captured by the high-speed camera in order to provide references for future boiling mechanism research.

The experimental results have shown the skillfully master of pool boiling experimental technology. In future work, the proposed approach is easily scalable to various material surfaces or different working fluids, for example, the graphene coating surface or nanofluid. The pool boiling experiment would be the benchmark data in the future experiments. Furthermore, the computer simulations would be validated by the experimental results.

In conclusion, it is believed that the boiling experiment in the current work is of great importance for the enhancement of efficiency and safety in the energy industry, particularly demonstrating the feasibility of developing a high-performance and durable heat exchange surface.

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Declaration of competing interest

The authors have no competing interests to declare that are relevant to the content of this article.

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